

DATA CLEANING & PREPARATION

Project 1 — Change Log
DecodeLabs Industrial Training Kit | Batch 2026

Audit Summary

Metric	Before Cleaning	After Cleaning
Total Rows	315	300
Duplicate Rows	10	0
Missing Values	81	0
Bad Date Formats	20	0
Negative Quantities	5	0
Price Outliers	3	0
Inconsistent Item Variants	23	10 (standardized)

Detailed Change Log

Change ID	Description	Impact	Status
CR001	Standardized Date format to ISO 8601 (YYYY-MM-DD)	Fixed 20 incorrectly formatted dates	Resolved
CR002	Imputed missing Quantity with median value	Preserved 16 records (median = 3)	Resolved
CR003	Replaced negative Quantity values	Fixed 5 invalid negative entries	Resolved
CR004	Capped price outliers & imputed nulls	3 outliers capped; 18 nulls filled via per-item median	Resolved
CR005	Imputed missing Payment_Method	Filled 20 nulls with mode value	Resolved
CR006	Removed duplicate rows	Dropped 10 duplicates; 305 unique records remain	Resolved
CR007	Standardized Item names (strip + title case)	Reduced 23 messy variants to 10 clean unique values	Resolved
CR008	Standardized Payment_Method casing	Strip + title case; 4 consistent values	Resolved

Change ID	Description	Impact	Status
CR009	Fixed 'NULL' string Category values	Corrected via Item-Category mapping	Resolved
CR010	Recalculated Total_Spent for all rows	Fixed 25 null totals; all rounded to 2dp	Resolved
CR011	Enforced correct data types	Date=datetime, Qty=int, prices=float	Resolved

VERIFICATION GATE — 0% Error Rate	
Unique Identifier Duplicates	0 ✓
Incorrectly Formatted Dates	0 ✓
Missing Values	0 ✓
Negative/Invalid Values	0 ✓

Dataset: Cafe Sales Dirty Data | Tool: Python (pandas) | DecodeLabs Batch 2026